
Problem Set 5

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PY 451 Section A1

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Problem 5

1.

$$\left(S_x^{(1)} \otimes S_y^{(2)}\right) (|+\rangle_1 \otimes |+\rangle_2) = \left(S_x^{(1)}|+\rangle_1\right) \otimes \left(S_y^{(2)}|+\rangle_2\right) = \frac{\hbar}{2}|- \rangle_1 \otimes \frac{i\hbar}{2}|- \rangle_2.$$

2. The norm squared is

$$\left(\frac{\hbar}{2}\langle -|_1 \otimes \frac{-i\hbar}{2}\langle -|_2\right) \left(\frac{\hbar}{2}| - \rangle_1 \otimes \frac{i\hbar}{2}| - \rangle_2\right) = \frac{\hbar^2}{4}\langle -| - \rangle_1 \cdot \frac{\hbar^2}{4}\langle -| - \rangle_2 = \frac{\hbar^4}{16}.$$

Therefore, the norm is $\frac{\hbar^2}{4}$.

Problem 2

Incase I forget to delete this before submitting the assignment, I included this section to justify my logic to myself so that after we had the lecture on tensor products, I could rewrite it in whatever presentation of the Born rule we gave in lecture.

The projector $P_a \otimes \mathbb{1}$ maps a two particle state $|\psi\rangle = |\psi\rangle_1 \otimes |\psi\rangle_2$ to the state $(P_a \otimes \mathbb{1})|\psi\rangle = |a\rangle_1 \otimes |\psi\rangle_2$. The probability of measuring the state $|a\rangle_1 \otimes |\psi\rangle_2$ in some system $|\phi\rangle$, that is, measuring $|a\rangle$ in the first particle and the same state $|\psi\rangle_2$ in the second, is given by the Born rule

$$|\langle\phi|(|a\rangle_1 \otimes |\psi\rangle_2)|^2.$$

This can be reframed as

$$|\langle\phi|(P_a \otimes \mathbb{1})|\psi\rangle|^2,$$

since

$$|\psi\rangle_1 \otimes |\psi\rangle_2 \xrightarrow{P_a \otimes \mathbb{1}} |a\rangle_1 \otimes |\psi\rangle_2.$$

1. Note that **NOTE THAT THERE IS EXTRA 2 ON LEFT**

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|+\rangle + |-\rangle)_1 \otimes 2|+\rangle_2 = |+\rangle_{x;1} \otimes 2|+\rangle_2.$$

$$\langle\psi|\psi\rangle = (\langle+_x|+_x\rangle)_1 (4\langle+|+\rangle)_2 = 4.$$

To normalize the state, by bilinearity¹ of the tensor product we have to choose a singular state to multiply the normalizing factor into. Since the second electron is not normalized, we redefine

$$|\psi\rangle = |+\rangle_{x;1} \otimes |+\rangle_2.$$

¹Meaning for any scalar c , $c(|x\rangle \otimes |y\rangle) = c|x\rangle \otimes |y\rangle = |x\rangle \otimes c|y\rangle$.

Measuring the first electron in $S_z \otimes 1$:

$$\begin{aligned}\mathcal{P}_+ &= \langle \psi | (P_+ \otimes \mathbb{1}) | \psi \rangle = \langle \psi | \left(|+\rangle\langle+|^{(1)} \otimes \mathbb{1}^{(2)} \right) | \psi \rangle \\ &= \langle \psi | (|+\rangle_1 \langle+| \psi \rangle_1 \otimes | \psi \rangle_2) \\ &= \langle \psi | + \rangle_1 \langle+| \psi \rangle_1 \cdot \langle \psi | \psi \rangle_2 \\ &= \frac{1}{\sqrt{2}^2} \cdot 1 = \frac{1}{2}.\end{aligned}$$

The normalized state is then simply $|+\rangle_1 \otimes |+\rangle_2$, since we have measured $|+\rangle_1$ and not modified the second electron state. We do not need to do the remainder of the calculations; the only other possible outcome of the measurement $S_z^{(1)}$ is $-\hbar/2$, and since the probabilities sum to 1, we must have that $\mathcal{P}_- = 1/2$ as well. Moreover, by the same logic, the new normalized state is $|-\rangle_1 \otimes |+\rangle_2$.

EVERYTHING ELSE IS WRONG

2. The state is not normalized, because the second electron state has a norm of 2:

$$(\langle+| + \langle-|)(|+\rangle + |-\rangle) = \langle+|+\rangle + \langle-|-\rangle = 2.$$

Similarly, the first electron state has a norm of $1/2$:

$$\frac{1}{4} (\langle+| + \langle-|)(|+\rangle + |-\rangle) = \frac{1}{4} (\langle+|+\rangle + \langle-|-\rangle) = \frac{1}{2}.$$

We need to normalize each individual particle state for the tensor to make physical sense. It is obvious that the normalized state is

$$|\psi\rangle = \frac{1}{\sqrt{2}} ((|+\rangle_1 + |-\rangle_1) \otimes (|+\rangle_2 + |-\rangle_2)) = |+\rangle_{x;1} \otimes |+\rangle_{x;2}.$$

We then have

$$\begin{aligned}\mathcal{P}_+ &= \langle \psi | \left(|+\rangle\langle+|^{(1)} \otimes \mathbb{1}^{(2)} \right) | \psi \rangle = \langle \psi | \left(|+\rangle_1 \langle+| \psi \rangle_1 \otimes \mathbb{1}^{(2)} | \psi \rangle_2 \right) \\ &= |\langle+| \psi \rangle_1|^2 \cdot \langle \psi | \psi \rangle_2 \\ &= |\langle+| + \rangle_{x;1}|^2 \\ &= \frac{1}{2}.\end{aligned}$$

By the same logic as the previous problem, our observed state is $|+\rangle_1 \otimes |+\rangle_{x;2}$. The probability $\mathcal{P}_-$ is also $1/2$, and the state would be $|-\rangle_1 \otimes |+\rangle_{x;2}$.

3. Note that

$$\begin{aligned}|\psi\rangle &= \frac{1}{\sqrt{2}} (|+\rangle_1 \otimes |+\rangle_2 + |-\rangle_1 \otimes |-\rangle_2) = \frac{1}{\sqrt{2}} ((|+\rangle_1 + |-\rangle_1) \otimes (|+\rangle_2 + |-\rangle_2)) \\ &= |+\rangle_{x;1} + \sqrt{2} |+\rangle_{x;2}.\end{aligned}$$

The last step follows by bilinearity of tensor products. We introduce a normalizing factor to the second electron state to obtain the state

$$|\psi\rangle = \frac{1}{\sqrt{2}} |+\rangle_{x;1} + \frac{1}{\sqrt{2}} |-\rangle_{x;2}.$$

This is the same state as part 2, so the answers are the same.

4. The calculations here are all basically the same as before, so I will start with the table, and will do the calculations afterwards. I will show less work because once again, they are basically the same as before.

	$\mathcal{P}\left(S_z^{(1)} = \frac{\hbar}{2}, S_z^{(2)} = \frac{\hbar}{2}\right)$	$\mathcal{P}\left(S_z^{(1)} = -\frac{\hbar}{2}, S_z^{(2)} = \frac{\hbar}{2}\right)$	$\mathcal{P}\left(S_z^{(1)} = \frac{\hbar}{2}, S_z^{(2)} = -\frac{\hbar}{2}\right)$	$\mathcal{P}\left(S_z^{(1)} = -\frac{\hbar}{2}, S_z^{(2)} = -\frac{\hbar}{2}\right)$
i	1/2	1/2	0	0
ii	1/4	1/4	1/4	1/4
ii	1/4	1/4	1/4	1/4

- (a) Recall the first normalized state after measurement was $|\psi\rangle = |\pm\rangle_1 \otimes |+\rangle_2$. Measuring again in $S_z^{(2)}$ gives

$$\begin{aligned}
 \mathcal{P}_{\pm,+} &= \langle\psi| \left(\mathbb{1}^{(1)} \otimes |+\rangle\langle+|^{(2)} \right) |\psi\rangle \\
 &= \langle\psi|\psi\rangle_1 \cdot |\langle\psi|+\rangle_2|^2 \\
 &= |\langle+|+\rangle_2|^2 \\
 &= 1.
 \end{aligned}$$

Multiplying this by the probability of already having measured a state $|+\rangle_1$ or $|-\rangle_1$ gives probabilities $\mathcal{P}_{+,+} = \mathcal{P}_{-,+} = 1/2$. We also find that the probability of measuring $|-\rangle$ is 0, since probabilities sum to 1.

- (b) The normalized post-measurement states were $|\pm\rangle_1 \otimes |+\rangle_{x;2}$. Measuring again in $S_z^{(2)}$ gives

$$\begin{aligned}
 \mathcal{P}_{\pm,+} &= \langle\psi| \left(\mathbb{1}^{(1)} \otimes |+\rangle\langle+|^{(2)} \right) |\psi\rangle \\
 &= \langle\psi|\psi\rangle \cdot |\langle\psi|+\rangle_2|^2 \\
 &= |\langle+_x|+\rangle|^2 \\
 &= \frac{1}{2}.
 \end{aligned}$$

Multiplying through by the probability of measuring either $|+\rangle_1$ or $|-\rangle_1$ gives probabilities of 1/4. Since probabilities sum to 1, we have $\mathcal{P}_{\pm,-} = 1/2$, and by the same logic we have all other probabilities are 1/4 as well.

- (c) Recall that the state here is the same as in problem 2. Therefore, the probabilities are the same as well.